

Homework №3

(**Deadline:** May 18, 2024)

1. (**2 points**) A pointlike nonrelativistic particle of mass m scatters in the time-dependent spherically symmetric Coulomb potential

$$U(r, t) = \frac{U_0(t)}{r}, \quad U_0(t) = \frac{\kappa}{\sqrt{t}}, \quad \kappa > 0. \quad (1)$$

- (a) (**0.5 point**) Identify all continuous symmetries of the action. Analyze if there are symmetries of type

$$\mathbf{r} \rightarrow \alpha \mathbf{r}, \quad t \rightarrow \beta t, \quad \alpha, \beta = \text{const}$$

which do not change the action or the equations of motion.

- (b) (**1 point**) Construct the integrals of motion which might exist according to Noether's theorem. Demonstrate that these quantities are conserved using the equations of motion.
- (c) (**0.5 point**) We have seen in lectures that in static (time-independent) Coulomb potential the particle's trajectory always remains in the plane (so-called scattering plane). Analyze if this property remains valid in the time-dependent potential given in Eq. (1). Analyze if the Kepler's laws remain valid in this potential field.
2. (**2 points**) A relativistic particle of mass m moves in the spherically symmetric potential $U(r) = kr^2/2$ of 3D harmonic oscillator.
- (a) (**0.5 point**) Write out the lagrangian of the system, and the corresponding equations of motion
- (b) (**0.5 point**) Identify all the symmetries of the action and construct all integrals of motion which might exist in this problem.
- (c) (**0.5 point**) Try to integrate the equations of motion and find trajectories for bound orbits ¹. Analyze if the trajectories are closed for bound orbits and if the motion remains periodic as a function of time.
- (d) (**0.5 point**) Demonstrate that in the limit $c \rightarrow \infty$ your results reproduce well-known nonrelativistic limit. Assuming that c is large yet finite, find explicitly the first relativistic correction to trajectory.

¹ **Help:** As usual, You may use integrals of motion instead of some or all equations of motion in order to find the trajectories.